

```
%pip install pandas matplotlib seaborn scikit-learn

# imports
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score
from sklearn.metrics import classification_report
from sklearn.metrics import confusion_matrix
from sklearn.metrics import roc_curve, roc_auc_score
```

Task 1: Data Acquisition and Exploration

1. Download and load the dataset

[Dataset Download link](#)

<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>

```
import pandas as pd

# Read wdbc.names file
with open("../Datasets/Lab
4/breast+cancer+wisconsin+diagnostic/wdbc.names", "r") as file:
    names_content = file.read()

print(names_content)
```

1. Title: Wisconsin Diagnostic Breast Cancer (WDBC)

2. Source Information

a) Creators:

Dr. William H. Wolberg, General Surgery Dept., University of Wisconsin, Clinical Sciences Center, Madison, WI 53792
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olvi@cs.wisc.edu

b) Donor: Nick Street

c) Date: November 1995

3. Past Usage:

first usage:

W.N. Street, W.H. Wolberg and O.L. Mangasarian
Nuclear feature extraction for breast tumor diagnosis.
IS&T/SPIE 1993 International Symposium on Electronic Imaging:
Science
and Technology, volume 1905, pages 861-870, San Jose, CA, 1993.

OR literature:

O.L. Mangasarian, W.N. Street and W.H. Wolberg.
Breast cancer diagnosis and prognosis via linear programming.
Operations Research, 43(4), pages 570-577, July-August 1995.

Medical literature:

W.H. Wolberg, W.N. Street, and O.L. Mangasarian.
Machine learning techniques to diagnose breast cancer from
fine-needle aspirates.
Cancer Letters 77 (1994) 163-171.

W.H. Wolberg, W.N. Street, and O.L. Mangasarian.
Image analysis and machine learning applied to breast cancer
diagnosis and prognosis.
Analytical and Quantitative Cytology and Histology, Vol. 17
No. 2, pages 77-87, April 1995.

W.H. Wolberg, W.N. Street, D.M. Heisey, and O.L. Mangasarian.
Computerized breast cancer diagnosis and prognosis from fine
needle aspirates.
Archives of Surgery 1995;130:511-516.

W.H. Wolberg, W.N. Street, D.M. Heisey, and O.L. Mangasarian.
Computer-derived nuclear features distinguish malignant from
benign breast cytology.
Human Pathology, 26:792--796, 1995.

See also:

<http://www.cs.wisc.edu/~olvi/uwmp/mpml.html>
<http://www.cs.wisc.edu/~olvi/uwmp/cancer.html>

Results:

- predicting field 2, diagnosis: B = benign, M = malignant
- sets are linearly separable using all 30 input features
- best predictive accuracy obtained using one separating plane

in the 3-D space of Worst Area, Worst Smoothness and Mean Texture. Estimated accuracy 97.5% using repeated 10-fold crossvalidations. Classifier has correctly diagnosed 176 consecutive new patients as of November 1995.

4. Relevant information

Features are computed from a digitized image of a fine needle aspirate (FNA) of a breast mass. They describe characteristics of the cell nuclei present in the image.

A few of the images can be found at
<http://www.cs.wisc.edu/~street/images/>

Separating plane described above was obtained using Multisurface Method-Tree (MSM-T) [K. P. Bennett, "Decision Tree Construction Via Linear Programming." Proceedings of the 4th Midwest Artificial Intelligence and Cognitive Science Society, pp. 97-101, 1992], a classification method which uses linear programming to construct a decision tree. Relevant features were selected using an exhaustive search in the space of 1-4 features and 1-3 separating planes.

The actual linear program used to obtain the separating plane in the 3-dimensional space is that described in:
[K. P. Bennett and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets", Optimization Methods and Software 1, 1992, 23-34].

This database is also available through the UW CS ftp server:

```
ftp ftp.cs.wisc.edu
cd math-prog/cpo-dataset/machine-learn/WDBC/
```

5. Number of instances: 569

6. Number of attributes: 32 (ID, diagnosis, 30 real-valued input features)

7. Attribute information

- 1) ID number
- 2) Diagnosis (M = malignant, B = benign)
- 3-32)

Ten real-valued features are computed for each cell nucleus:

- a) radius (mean of distances from center to points on the perimeter)

- b) texture (standard deviation of gray-scale values)
- c) perimeter
- d) area
- e) smoothness (local variation in radius lengths)
- f) compactness ($\text{perimeter}^2 / \text{area} - 1.0$)
- g) concavity (severity of concave portions of the contour)
- h) concave points (number of concave portions of the contour)
- i) symmetry
- j) fractal dimension ("coastline approximation" - 1)

Several of the papers listed above contain detailed descriptions of how these features are computed.

The mean, standard error, and "worst" or largest (mean of the three largest values) of these features were computed for each image, resulting in 30 features. For instance, field 3 is Mean Radius, field 13 is Radius SE, field 23 is Worst Radius.

All feature values are recoded with four significant digits.

8. Missing attribute values: none

9. Class distribution: 357 benign, 212 malignant

```
# import pandas as pd
```

```
df = pd.read_csv('../Datasets/Lab
4/breast+cancer+wisconsin+diagnostic/wdbc.data', header=None)
df.head()
```

	0	1	2	3	4	5	6	7	8
\									
0	842302	M	17.99	10.38	122.80	1001.0	0.11840	0.27760	0.3001
1	842517	M	20.57	17.77	132.90	1326.0	0.08474	0.07864	0.0869
2	84300903	M	19.69	21.25	130.00	1203.0	0.10960	0.15990	0.1974
3	84348301	M	11.42	20.38	77.58	386.1	0.14250	0.28390	0.2414
4	84358402	M	20.29	14.34	135.10	1297.0	0.10030	0.13280	0.1980
	9	...	22	23	24	25	26	27	28
29 \									
0	0.14710	...	25.38	17.33	184.60	2019.0	0.1622	0.6656	0.7119
	0.2654								
1	0.07017	...	24.99	23.41	158.80	1956.0	0.1238	0.1866	0.2416
	0.1860								
2	0.12790	...	23.57	25.53	152.50	1709.0	0.1444	0.4245	0.4504
	0.2430								

```

3  0.10520  ...  14.91  26.50   98.87   567.7   0.2098   0.8663   0.6869
0.2575
4  0.10430  ...  22.54  16.67  152.20  1575.0   0.1374   0.2050   0.4000
0.1625

```

```

      30      31
0  0.4601  0.11890
1  0.2750  0.08902
2  0.3613  0.08758
3  0.6638  0.17300
4  0.2364  0.07678

```

```
[5 rows x 32 columns]
```

```
# import pandas as pd
```

```
# Column names according to wdbc.names
```

```

columns = [
    "ID", "Diagnosis",
    "radius_mean", "texture_mean", "perimeter_mean", "area_mean",
    "smoothness_mean", "compactness_mean", "concavity_mean",
    "concave_points_mean", "symmetry_mean", "fractal_dimension_mean",

    "radius_se", "texture_se", "perimeter_se", "area_se",
    "smoothness_se", "compactness_se", "concavity_se",
    "concave_points_se", "symmetry_se", "fractal_dimension_se",

    "radius_worst", "texture_worst", "perimeter_worst", "area_worst",
    "smoothness_worst", "compactness_worst", "concavity_worst",
    "concave_points_worst", "symmetry_worst",
    "fractal_dimension_worst"
]

```

```
# Load dataset
```

```

df = pd.read_csv("../Datasets/Lab
4/breast+cancer+wisconsin+diagnostic/wdbc.data", header=None,
names=columns)

```

```
# Display first 5 rows
```

```
df.head()
```

	ID	Diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
0	842302	M	17.99	10.38	122.80	1001.0
1	842517	M	20.57	17.77	132.90	1326.0
2	84300903	M	19.69	21.25	130.00	1203.0
3	84348301	M	11.42	20.38	77.58	

```

386.1
4 84358402 M 20.29 14.34 135.10
1297.0

    smoothness_mean compactness_mean concavity_mean
concave_points_mean \
0 0.11840 0.27760 0.3001
0.14710
1 0.08474 0.07864 0.0869
0.07017
2 0.10960 0.15990 0.1974
0.12790
3 0.14250 0.28390 0.2414
0.10520
4 0.10030 0.13280 0.1980
0.10430

    ... radius_worst texture_worst perimeter_worst area_worst \
0 ... 25.38 17.33 184.60 2019.0
1 ... 24.99 23.41 158.80 1956.0
2 ... 23.57 25.53 152.50 1709.0
3 ... 14.91 26.50 98.87 567.7
4 ... 22.54 16.67 152.20 1575.0

    smoothness_worst compactness_worst concavity_worst
concave_points_worst \
0 0.1622 0.6656 0.7119
0.2654
1 0.1238 0.1866 0.2416
0.1860
2 0.1444 0.4245 0.4504
0.2430
3 0.2098 0.8663 0.6869
0.2575
4 0.1374 0.2050 0.4000
0.1625

    symmetry_worst fractal_dimension_worst
0 0.4601 0.11890
1 0.2750 0.08902
2 0.3613 0.08758
3 0.6638 0.17300
4 0.2364 0.07678

[5 rows x 32 columns]

```

1. Compute summary statistics (mean, median, standard deviation, etc.) for all features.

```

# Summary statistics for numerical features
df.describe()

```

	ID	radius_mean	texture_mean	perimeter_mean
area_mean \				
count	5.690000e+02	569.000000	569.000000	569.000000
mean	3.037183e+07	14.127292	19.289649	91.969033
std	1.250206e+08	3.524049	4.301036	24.298981
min	8.670000e+03	6.981000	9.710000	43.790000
25%	8.692180e+05	11.700000	16.170000	75.170000
50%	9.060240e+05	13.370000	18.840000	86.240000
75%	8.813129e+06	15.780000	21.800000	104.100000
max	9.113205e+08	28.110000	39.280000	188.500000

	smoothness_mean	compactness_mean	concavity_mean
concave_points_mean \			
count	569.000000	569.000000	569.000000
mean	0.096360	0.104341	0.088799
std	0.014064	0.052813	0.079720
min	0.052630	0.019380	0.000000
25%	0.086370	0.064920	0.029560
50%	0.095870	0.092630	0.061540
75%	0.105300	0.130400	0.130700
max	0.163400	0.345400	0.426800

	symmetry_mean	...	radius_worst	texture_worst
perimeter_worst \				
count	569.000000	...	569.000000	569.000000
mean	0.181162	...	16.269190	25.677223
std	0.027414	...	4.833242	6.146258
min	0.106000	...	7.930000	12.020000
25%	0.161900	...	13.010000	21.080000

50%	0.179200	...	14.970000	25.410000
97.660000				
75%	0.195700	...	18.790000	29.720000
125.400000				
max	0.304000	...	36.040000	49.540000
251.200000				

	area_worst	smoothness_worst	compactness_worst
concavity_worst \			
count	569.000000	569.000000	569.000000
569.000000			
mean	880.583128	0.132369	0.254265
0.272188			
std	569.356993	0.022832	0.157336
0.208624			
min	185.200000	0.071170	0.027290
0.000000			
25%	515.300000	0.116600	0.147200
0.114500			
50%	686.500000	0.131300	0.211900
0.226700			
75%	1084.000000	0.146000	0.339100
0.382900			
max	4254.000000	0.222600	1.058000
1.252000			

	concave_points_worst	symmetry_worst	fractal_dimension_worst
count	569.000000	569.000000	569.000000
mean	0.114606	0.290076	0.083946
std	0.065732	0.061867	0.018061
min	0.000000	0.156500	0.055040
25%	0.064930	0.250400	0.071460
50%	0.099930	0.282200	0.080040
75%	0.161400	0.317900	0.092080
max	0.291000	0.663800	0.207500

[8 rows x 31 columns]

```
summary = pd.DataFrame({
    "Mean": df.mean(numeric_only=True),
    "Median": df.median(numeric_only=True),
    "Standard Deviation": df.std(numeric_only=True)
})
```

summary

	Mean	Median	Standard
Deviation			
ID	3.037183e+07	906024.000000	
1.250206e+08			

radius_mean	1.412729e+01	13.370000	
3.524049e+00			
texture_mean	1.928965e+01	18.840000	
4.301036e+00			
perimeter_mean	9.196903e+01	86.240000	
2.429898e+01			
area_mean	6.548891e+02	551.100000	
3.519141e+02			
smoothness_mean	9.636028e-02	0.095870	1.406413e-
02			
compactness_mean	1.043410e-01	0.092630	5.281276e-
02			
concavity_mean	8.879932e-02	0.061540	7.971981e-
02			
concave_points_mean	4.891915e-02	0.033500	3.880284e-
02			
symmetry_mean	1.811619e-01	0.179200	2.741428e-
02			
fractal_dimension_mean	6.279761e-02	0.061540	7.060363e-
03			
radius_se	4.051721e-01	0.324200	2.773127e-
01			
texture_se	1.216853e+00	1.108000	5.516484e-
01			
perimeter_se	2.866059e+00	2.287000	
2.021855e+00			
area_se	4.033708e+01	24.530000	
4.549101e+01			
smoothness_se	7.040979e-03	0.006380	3.002518e-
03			
compactness_se	2.547814e-02	0.020450	1.790818e-
02			
concavity_se	3.189372e-02	0.025890	3.018606e-
02			
concave_points_se	1.179614e-02	0.010930	6.170285e-
03			
symmetry_se	2.054230e-02	0.018730	8.266372e-
03			
fractal_dimension_se	3.794904e-03	0.003187	2.646071e-
03			
radius_worst	1.626919e+01	14.970000	
4.833242e+00			
texture_worst	2.567722e+01	25.410000	
6.146258e+00			
perimeter_worst	1.072612e+02	97.660000	
3.360254e+01			
area_worst	8.805831e+02	686.500000	
5.693570e+02			
smoothness_worst	1.323686e-01	0.131300	2.283243e-

02			
compactness_worst	2.542650e-01	0.211900	1.573365e-
01			
concavity_worst	2.721885e-01	0.226700	2.086243e-
01			
concave_points_worst	1.146062e-01	0.099930	6.573234e-
02			
symmetry_worst	2.900756e-01	0.282200	6.186747e-
02			
fractal_dimension_worst	8.394582e-02	0.080040	1.806127e-
02			

1. Check for missing values, duplicates, or outliers. Handle them

```
# Check missing values in each column
missing_values = df.isnull().sum()

print("Missing values in each column:")
print(missing_values)

print("\nTotal missing values:", missing_values.sum())
```

Missing values in each column:

ID	0
Diagnosis	0
radius_mean	0
texture_mean	0
perimeter_mean	0
area_mean	0
smoothness_mean	0
compactness_mean	0
concavity_mean	0
concave_points_mean	0
symmetry_mean	0
fractal_dimension_mean	0
radius_se	0
texture_se	0
perimeter_se	0
area_se	0
smoothness_se	0
compactness_se	0
concavity_se	0
concave_points_se	0
symmetry_se	0
fractal_dimension_se	0
radius_worst	0
texture_worst	0
perimeter_worst	0
area_worst	0
smoothness_worst	0

```
compactness_worst      0
concavity_worst        0
concave_points_worst   0
symmetry_worst         0
fractal_dimension_worst 0
dtype: int64
```

Total missing values: 0

```
# Check duplicate rows
```

```
duplicates = df.duplicated().sum()
```

```
print("Number of duplicate rows:", duplicates)
```

Number of duplicate rows: 0

```
# Select the 30 numerical features
```

```
features = df.columns[2:]
```

```
# Calculate Q1, Q3 and IQR
```

```
Q1 = df[features].quantile(0.25)
```

```
Q3 = df[features].quantile(0.75)
```

```
IQR = Q3 - Q1
```

```
# Identify outliers
```

```
outliers = ((df[features] < (Q1 - 1.5 * IQR)) |
            (df[features] > (Q3 + 1.5 * IQR)))
```

```
# Count outliers in each feature
```

```
outlier_counts = outliers.sum()
```

```
print("Number of outliers in each feature:")
```

```
print(outlier_counts)
```

Number of outliers in each feature:

```
radius_mean      14
texture_mean     7
perimeter_mean  13
area_mean       25
smoothness_mean  6
compactness_mean 16
concavity_mean   18
concave_points_mean 10
symmetry_mean    15
fractal_dimension_mean 15
radius_se       38
texture_se      20
perimeter_se    38
area_se        65
smoothness_se   30
compactness_se  28
```

```

concavity_se          22
concave_points_se     19
symmetry_se           27
fractal_dimension_se  28
radius_worst          17
texture_worst         5
perimeter_worst       15
area_worst            35
smoothness_worst      7
compactness_worst     16
concavity_worst       12
concave_points_worst  0
symmetry_worst        23
fractal_dimension_worst 24
dtype: int64

```

Handle outliers using IQR capping

```
lower_bound = Q1 - 1.5 * IQR
```

```
upper_bound = Q3 + 1.5 * IQR
```

```

df[features] = df[features].clip(
    lower=lower_bound,
    upper=upper_bound,
    axis=1
)

```

```
print("Outliers handled using IQR capping.")
```

Outliers handled using IQR capping.

```
print("Final dataset shape:", df.shape)
```

```
print("Total missing values:", df.isnull().sum().sum())
```

```
print("Total duplicate rows:", df.duplicated().sum())
```

Final dataset shape: (569, 32)

Total missing values: 0

Total duplicate rows: 0

1. Visualize the data: Create histograms or box plots for key features, a correlation heatmap, and class distribution.

```
# import matplotlib.pyplot as plt
```

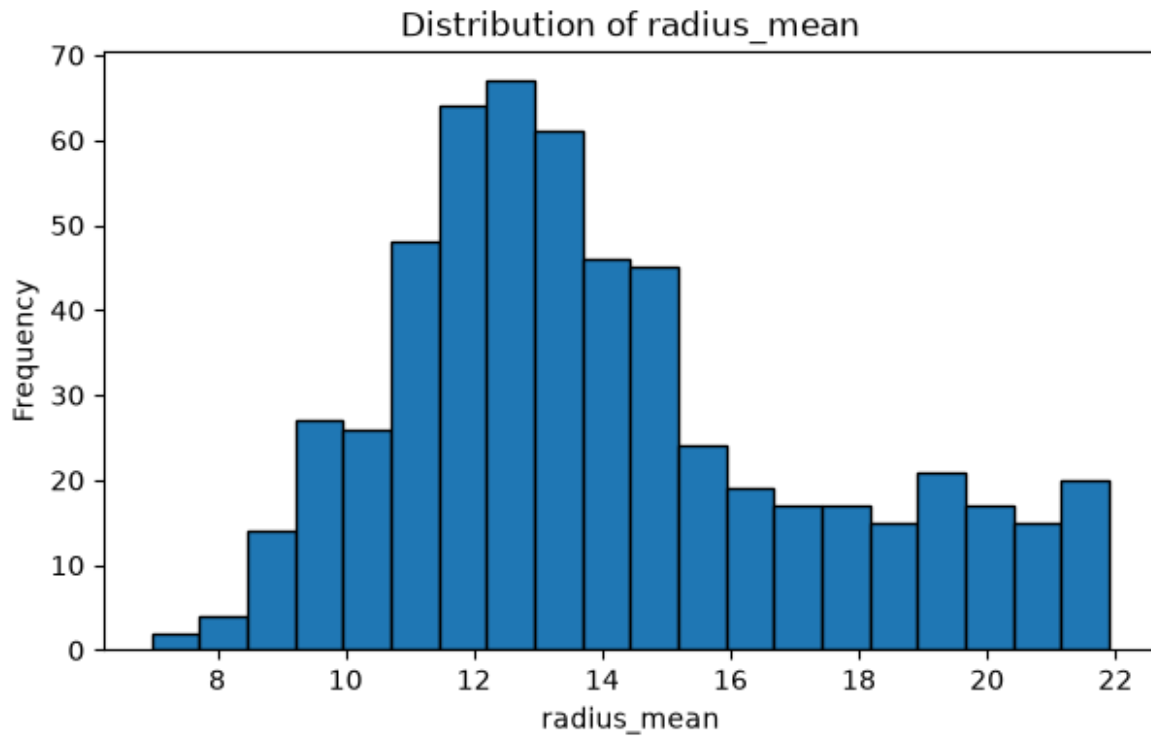
Key features

```

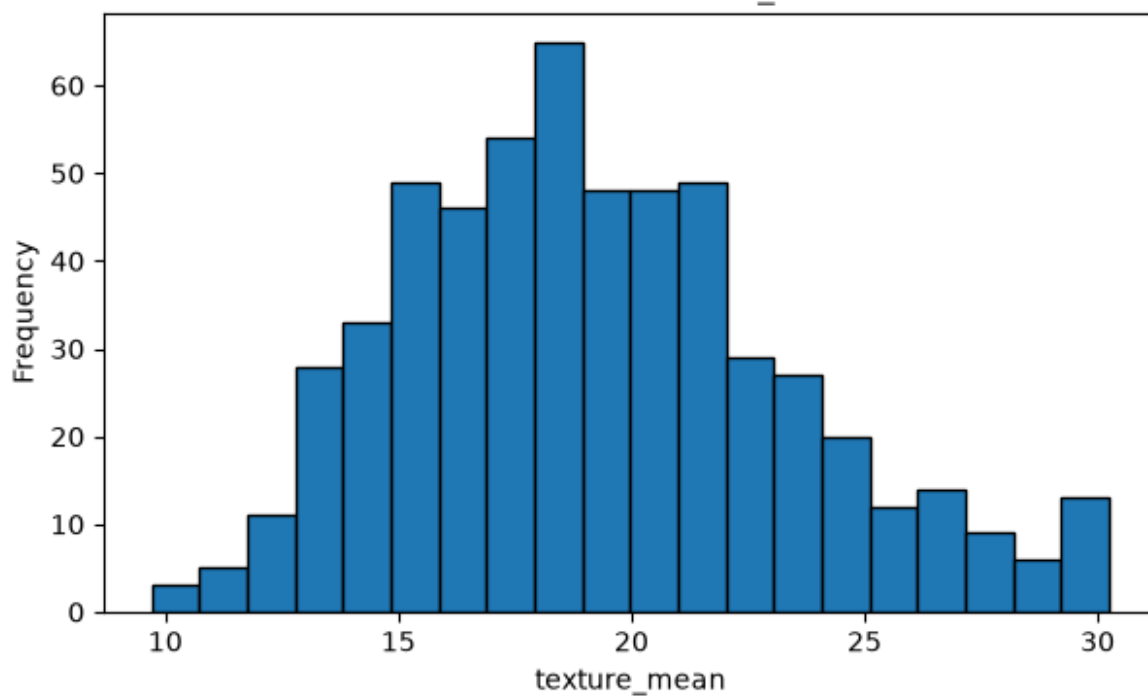
key_features = [
    "radius_mean",
    "texture_mean",
    "perimeter_mean",
    "area_mean",
    "smoothness_mean"
]

```

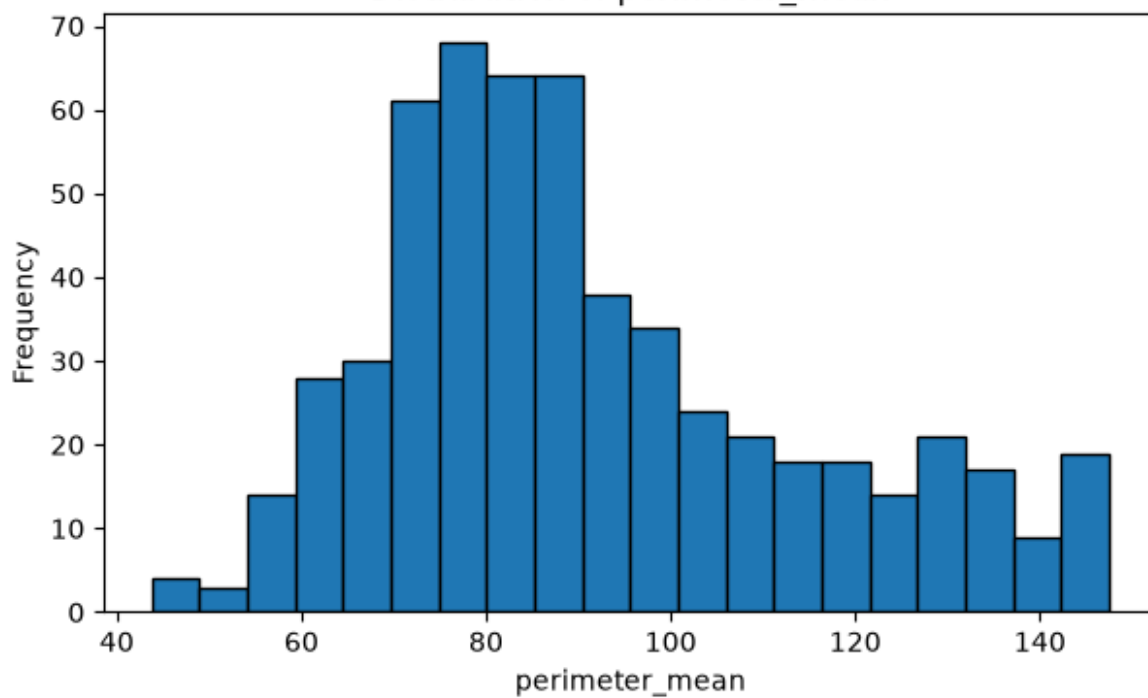
```
# Create histograms
for feature in key_features:
    plt.figure(figsize=(7, 4))
    plt.hist(df[feature], bins=20, edgecolor="black")
    plt.title(f"Distribution of {feature}")
    plt.xlabel(feature)
    plt.ylabel("Frequency")
    plt.show()
```

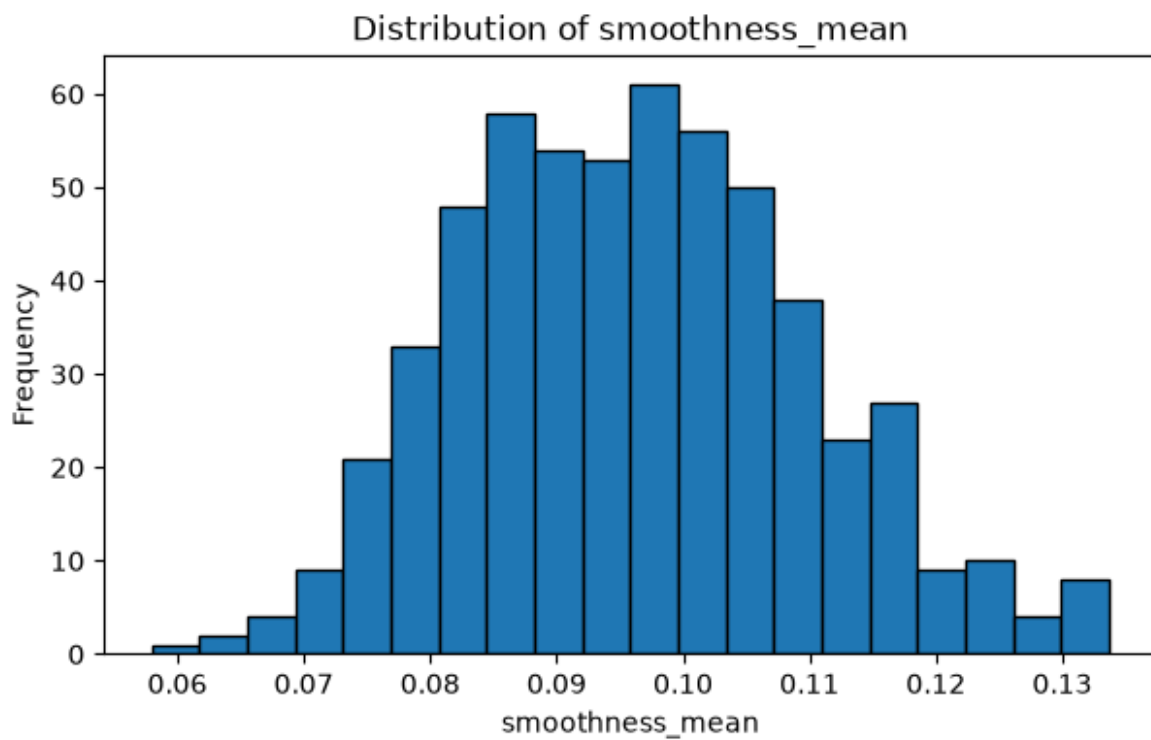
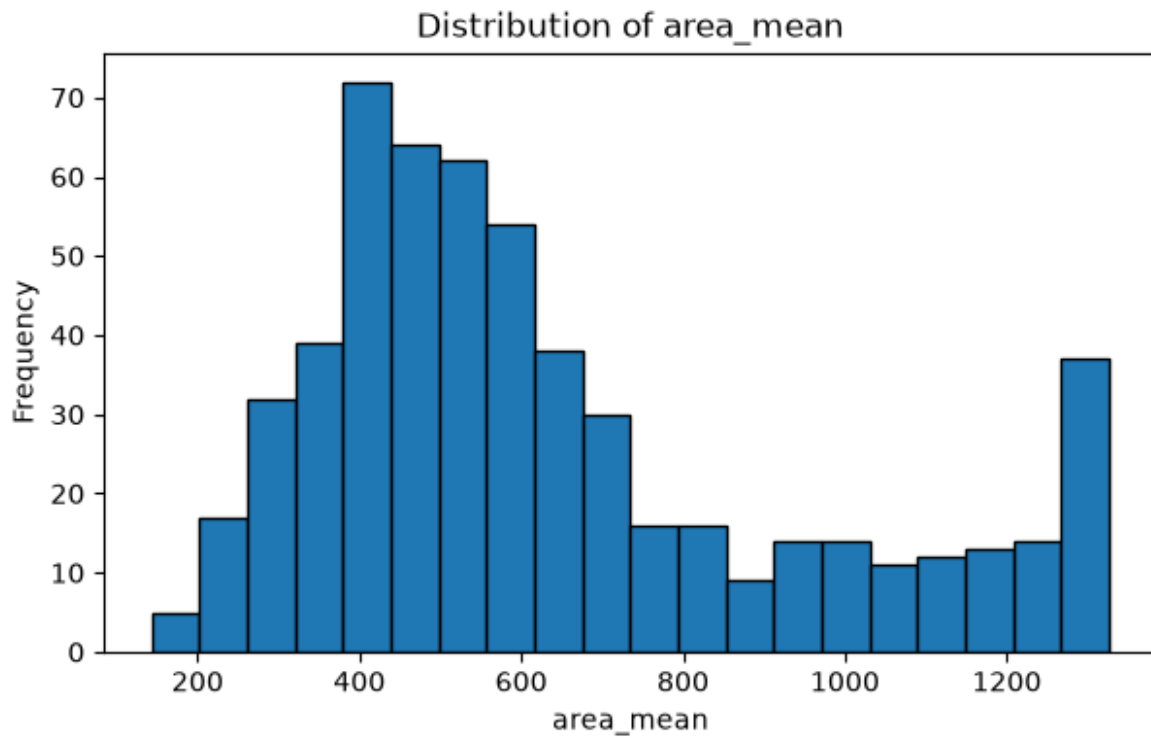


Distribution of texture_mean



Distribution of perimeter_mean





```
# import seaborn as sns  
# import matplotlib.pyplot as plt
```

```

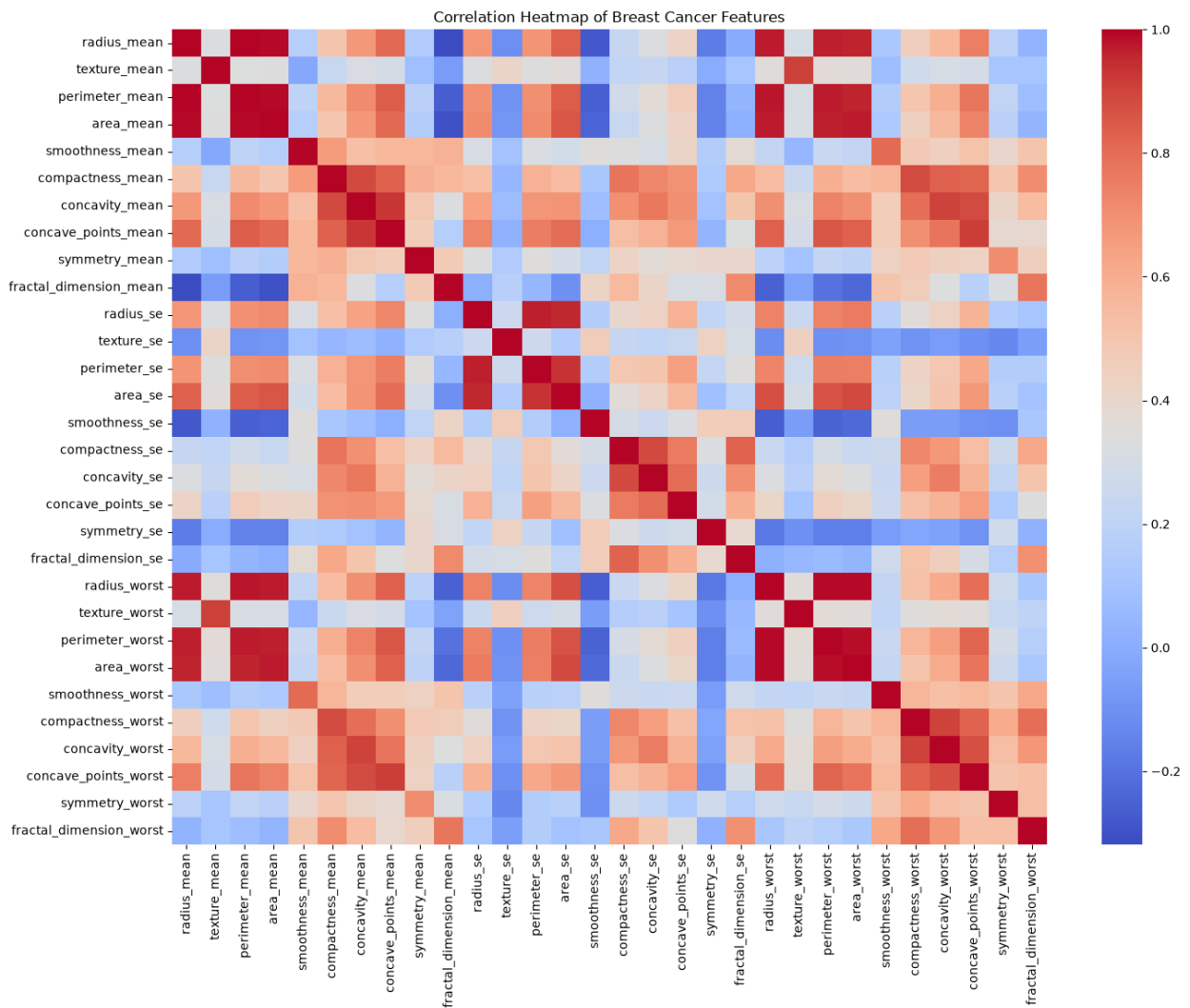
# Select numerical features
numerical_features = df.iloc[:, 2:]

# Calculate correlation matrix
correlation_matrix = numerical_features.corr()

# Plot heatmap
plt.figure(figsize=(16, 12))
sns.heatmap(
    correlation_matrix,
    cmap="coolwarm",
    annot=False
)

plt.title("Correlation Heatmap of Breast Cancer Features")
plt.show()

```

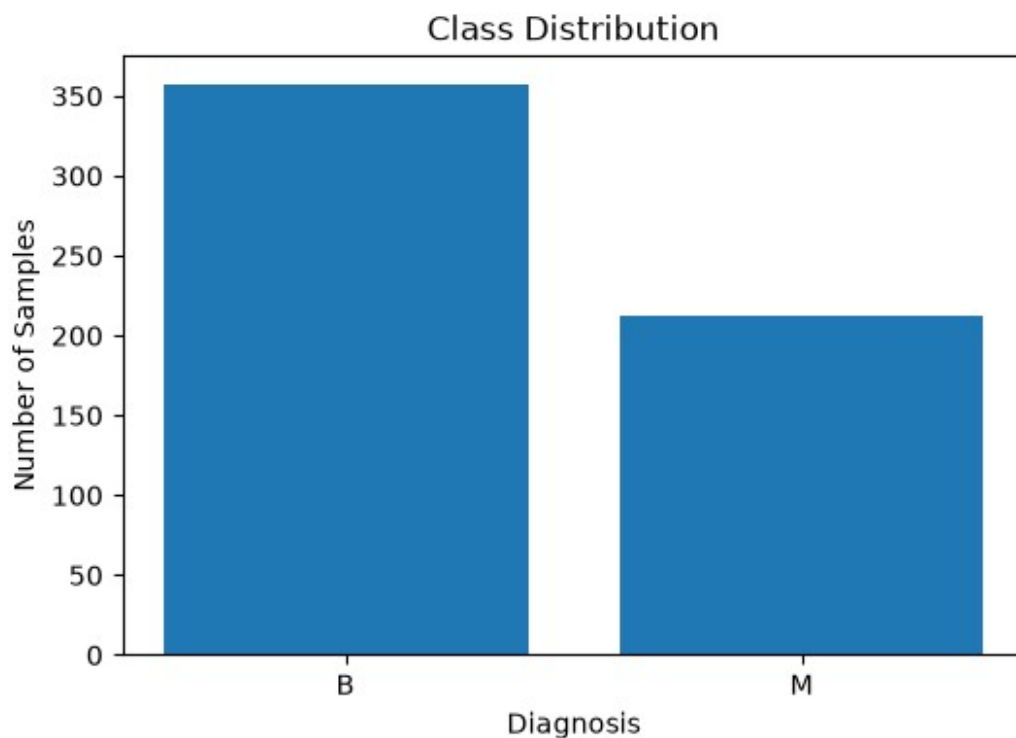


```
# Count the classes
class_counts = df["Diagnosis"].value_counts()

# Plot class distribution
plt.figure(figsize=(6, 4))
plt.bar(class_counts.index, class_counts.values)

plt.title("Class Distribution")
plt.xlabel("Diagnosis")
plt.ylabel("Number of Samples")

plt.show()
```



1. Discuss any imbalances in the class distribution and potential implications for model training.

This dataset has moderate class imbalance. This imbalance may cause the logistic regression model to favor the majority class and may reduce its ability to correctly identify malignant cases.

Task 2: Data Preprocessing

- Encode the target variable (Outcome) as binary

```
# Encode Diagnosis as binary
df["Diagnosis"] = df["Diagnosis"].map({"B": 0, "M": 1})
```

```
# Check the encoded values
print(df["Diagnosis"].value_counts())
```

```
Diagnosis
0      357
1      212
Name: count, dtype: int64
```

- Split the data into training and testing sets (e.g., 80/20 split) using stratified sampling to maintain class balance.

```
# from sklearn.model_selection import train_test_split

# Features and target
X = df.drop(columns=["ID", "Diagnosis"])
y = df["Diagnosis"]

# Split data into training and testing sets (80/20)
# Stratify ensures the same class proportion in both sets
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)
```

```
# Check the shapes
print("Training set shape:", X_train.shape)
print("Testing set shape:", X_test.shape)
```

```
Training set shape: (455, 30)
Testing set shape: (114, 30)
```

This is done using stratify to handle class imbalance and now we are check that class imbalance is handled successfully or not.

```
print("Training set class percentage:")
print(y_train.value_counts(normalize=True) * 100)

print("\nTesting set class percentage:")
print(y_test.value_counts(normalize=True) * 100)
```

```
Training set class percentage:
Diagnosis
0      62.637363
1      37.362637
Name: proportion, dtype: float64
```

```
Testing set class percentage:
```

```
Diagnosis
0      63.157895
1      36.842105
Name: proportion, dtype: float64
```

- Standardize the features to ensure they are on the same scale, as logistic regression is sensitive to feature scales.

```
# from sklearn.preprocessing import StandardScaler

# Create StandardScaler
scaler = StandardScaler()

# Fit on training data and transform training data
X_train_scaled = scaler.fit_transform(X_train)

# Transform test data using the same scaler
X_test_scaled = scaler.transform(X_test)

# Check the shapes
print("Scaled training set shape:", X_train_scaled.shape)
print("Scaled testing set shape:", X_test_scaled.shape)

Scaled training set shape: (455, 30)
Scaled testing set shape: (114, 30)
```

Task 3: Model Training and Basic Evaluation

- Train a logistic regression model

```
# from sklearn.linear_model import LogisticRegression

# Create Logistic Regression model
logistic_model = LogisticRegression(random_state=42)

# Train the model
logistic_model.fit(X_train_scaled, y_train)

print("Logistic Regression model trained successfully.")

Logistic Regression model trained successfully.
```

- Evaluate the model on the test set using the following metrics:
- Accuracy
- Precision, Recall, and F1-score (report for both classes)
- Confusion matrix (visualize it)
- ROC curve and AUC score

```
# from sklearn.metrics import accuracy_score
y_pred = logistic_model.predict(X_test_scaled)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
Accuracy: 0.9649122807017544
# from sklearn.metrics import classification_report
print(classification_report(
    y_test,
    y_pred,
    target_names=["Benign", "Malignant"]
))
```

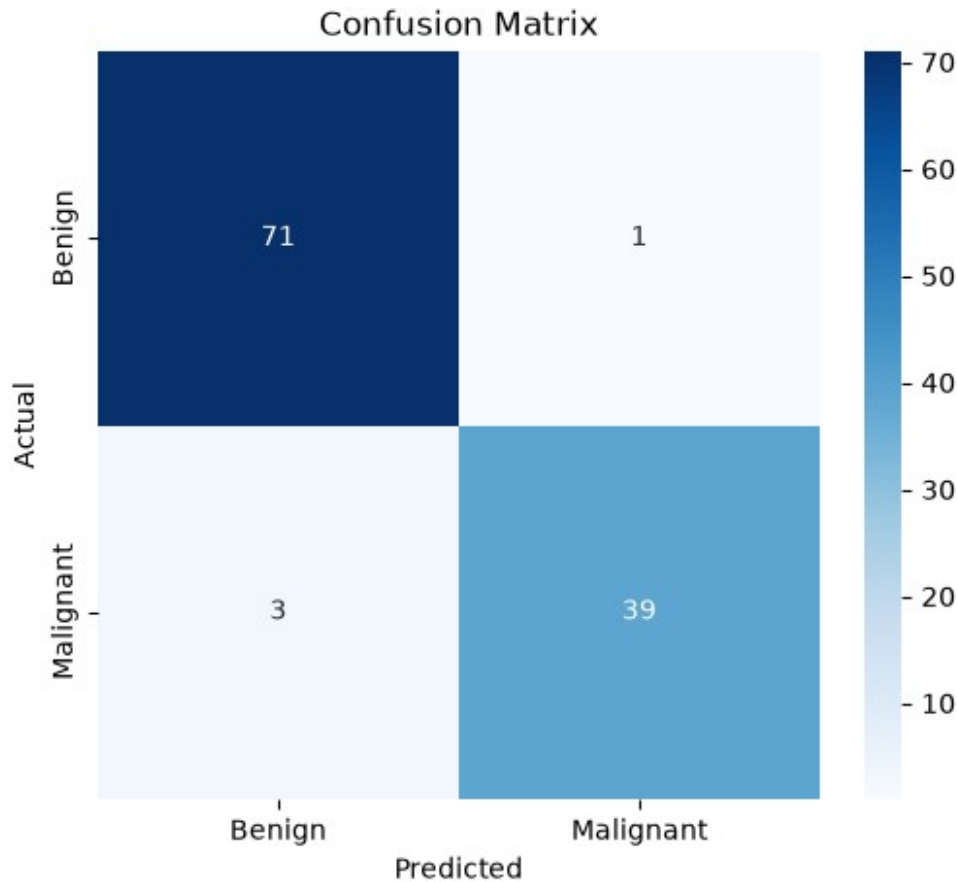
	precision	recall	f1-score	support
Benign	0.96	0.99	0.97	72
Malignant	0.97	0.93	0.95	42
accuracy			0.96	114
macro avg	0.97	0.96	0.96	114
weighted avg	0.97	0.96	0.96	114

```
# from sklearn.metrics import confusion_matrix
# import seaborn as sns
# import matplotlib.pyplot as plt
```

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6, 5))
sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["Benign", "Malignant"],
    yticklabels=["Benign", "Malignant"]
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
# from sklearn.metrics import roc_curve, roc_auc_score
# import matplotlib.pyplot as plt

y_prob = logistic_model.predict_proba(X_test_scaled)[: , 1]

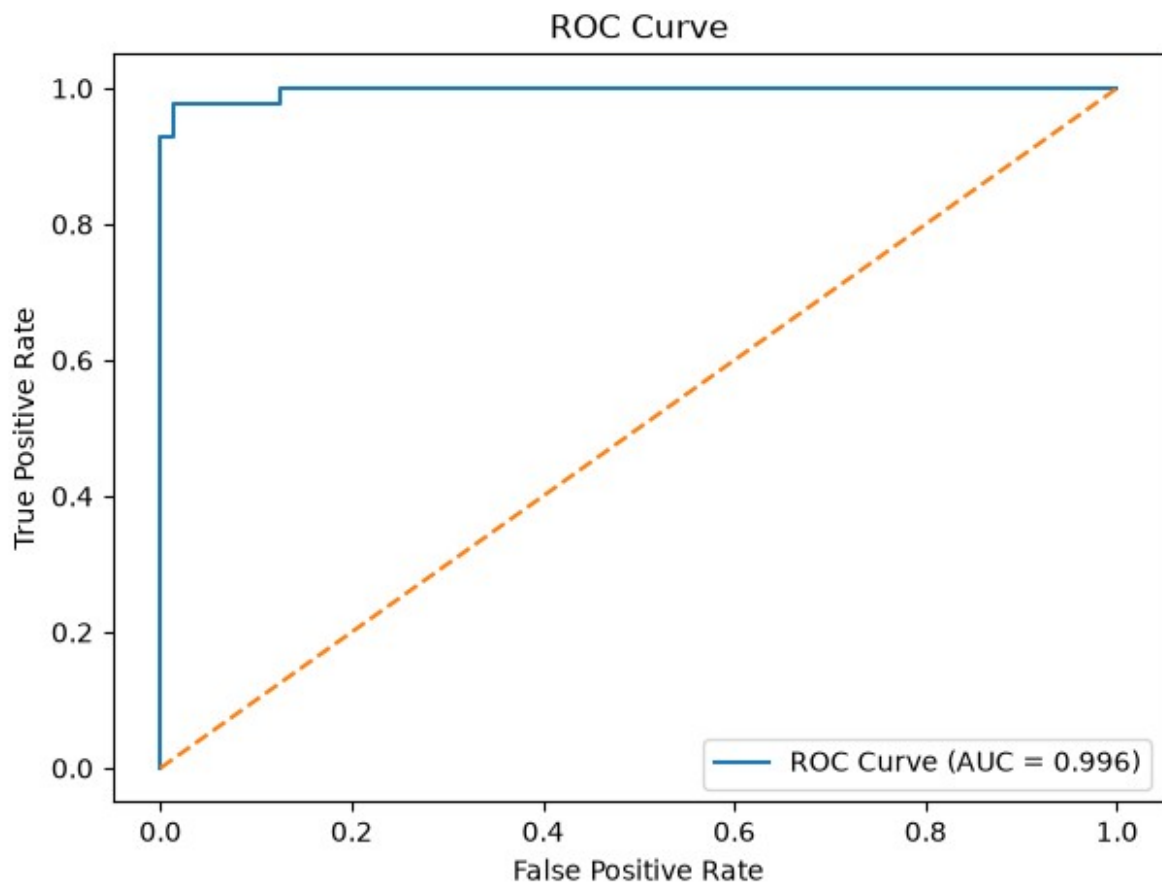
fpr, tpr, thresholds = roc_curve(y_test, y_prob)

auc_score = roc_auc_score(y_test, y_prob)

plt.figure(figsize=(7, 5))
plt.plot(fpr, tpr, label=f"ROC Curve (AUC = {auc_score:.3f})")
plt.plot([0, 1], [0, 1], linestyle="--")

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

print("AUC Score:", auc_score)
```



AUC Score: 0.9963624338624338